Introduction

Breast cancer is a prevalent issue in the world which requires a lot of resources and research put into it to decrease mortality rates. Notably, the survival rate fluctuates drastically depending on the spread of cancer before it is diagnosed, emphasizing the need to detect cancer as early as possible (United States Cancer Statistics, 2024).

Exploring whether a linear model where compactness is an independent variable helping us to predict concavity of the cell might provide a foundation for a lot of the potential diagnostics and evaluation tools. If there's a strong linear relationship, compactness could serve as a simple and accessible feature to approximate cell concavity. Compactness is easier to measure than concavity because it relies on perimeter and area, which are geometric properties that can be more easily extracted from the imaging.

Additionally, outside research shows that breast cells with more curvature, irregular forms (higher concavity) are more concerning as they are more likely to be cancerous (National Library of Medicine, 2021). So being able to predict those with higher concavity using compactness might be helpful in continuation of decreasing mortality rates.

To guide the study, the following research question is considered: “Can a linear model predict breast cell concavity based on its compactness?”
The population that we are intending to make the inference about is all breast masses, both malignant and benign. Our specific interest lies within measures of compactness and concavity.

Data analysis

The Breast Cancer Wisconsin Data Set contains 569 instances of breast cancer samples, classified as either benign or malignant. It includes 30 features related to the cell characteristics derived from digitized images of fine needle aspirates (Breast Cancer Wisconsin (Diagnostic), 1995).

For the purpose of this study, I will be working with two variables: concavity and compactness, which can be found in corresponding columns named “concavity_mean” and “compactness_mean”. None of those columns contained unknown values. See Appendix A for code.

Concavity:

- Description: Concavity measures the quality of curving in. It measures how deeply the boundaries of the cell curve inward, which can indicate more irregular shapes. In this particular dataset, concavity has a range of [0, 1], where values closer to 1 indicate more profound curvature.
- Type: Continuous quantitative variable. Concavity can take any real value within a specific range.
- Units: Dimensionless
- How measured: Concavity in fine needle aspirate (FNA) imaging is measured by analyzing how deeply the edges of a cell mass curve inward along its boundary, using contour detection techniques on the digitized image (Statistical Inference, 2024) .
- Importance: Higher concavity values typically indicate a more irregular and potentially malignant mass. In my study, this variable is **dependent**.

Compactness:

- Description: Compactness measures how compact or "tight" the shape of a breast mass is. It is calculated using the formula (Breast Cancer Wisconsin (Diagnostic), 1995):

$$Compactness = \frac{Perimeter^2}{Area} - 1$$

  This formula reflects how much the perimeter of the cell mass deviates from the area, which is an indicator of irregularity in shape. The higher the compactness value, the more irregular and less compact the shape of the mass.

- Type: Continuous quantitative variable. Compactness variable can take any real value within a certain range, making the possibility of the values infinite.
- Units: Dimensionless
- How measured: Perimeter and area, both of which are extracted from the digitized image of a fine needle aspirate (FNA) of the breast mass and after the above formula is used to calculate value.
- Importance: Compactness is easier to measure compared to concavity so it can serve as a good **predictor/independent** variable.

Descriptive Statistics

Key statistics needed from both variables to perform the analysis are mean and standard deviation as we need it for all the further calculations. Also, other relevant statistics were examined such as skewness as it will allow us to describe the distribution better and draw in conclusion how those can affect a future model.

The descriptive statistics of variables are calculated in Appendix B. The relevant summary statistics are displayed in Table 1.

Table 1: Summary statistics for the cell compactness (predictor variable - X) and cell concavity (response variable - Y). See Appendix B for calculations.

	Concavity	Compactness
Count	$n = 569$	
Mean	$\bar{y} = 0.08879931581722$	$\bar{x} = 0.10434098418277$
Median	0.06154	0.09263
Mode	0	$[0.1206, 0.1147]$ *(both values have the same frequency - the highest)*

Standard Deviation	$s_y = 0.07971980870789$	$s_x = 0.05281275793251$
Range	0.4268	0.32602
Skewness	1.4011797389486706	1.1901230311980362

The mean, median, and mode for both variables indicate a right-skewness. Concavity shows higher skewness (1.40) compared to compactness (1.19), which may affect linearity in regression. This high skewness may affect the assumptions of linear regression, especially normality and homoscedasticity. Skewed data could lead to non-normal residuals and unequal variance.

Correlation and Regression

Before proceeding with linear regression, we should examine Pearson's r which will show us what the strength and direction of this linear relationship is. Scatter plot (Figure 1) will be used to connect the trend observed. To calculate Pearson's R python was used. Calculations can be seen in Appendix C.

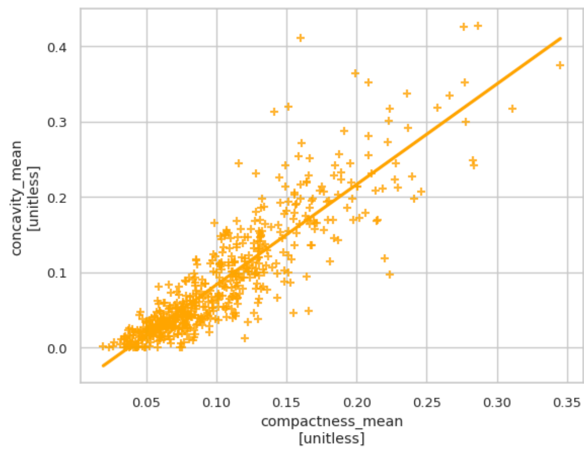


Figure 1: Scatter plot of compactness versus concavity

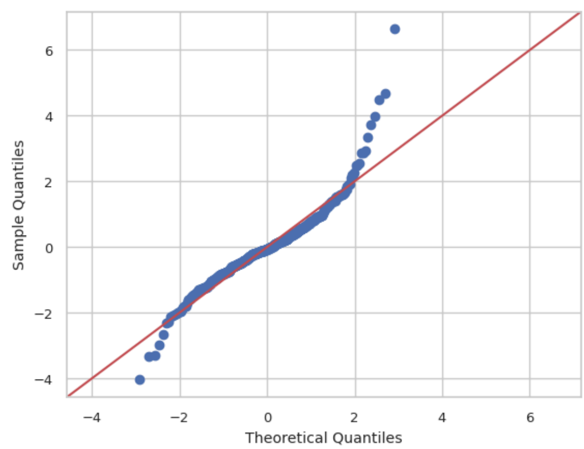


Figure 2: Normality (QQ) plot for residuals

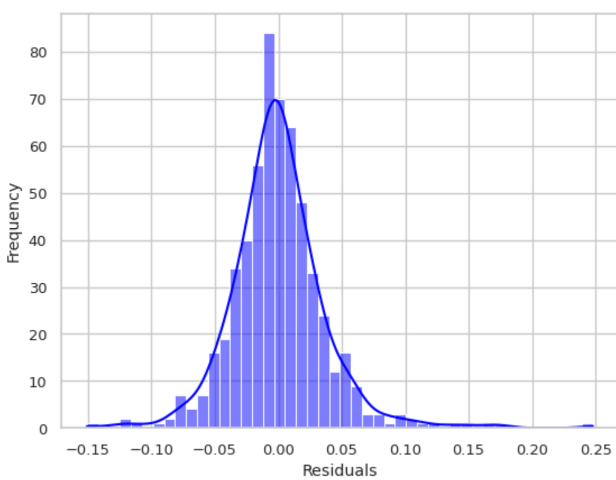


Figure 3: Histogram of residuals

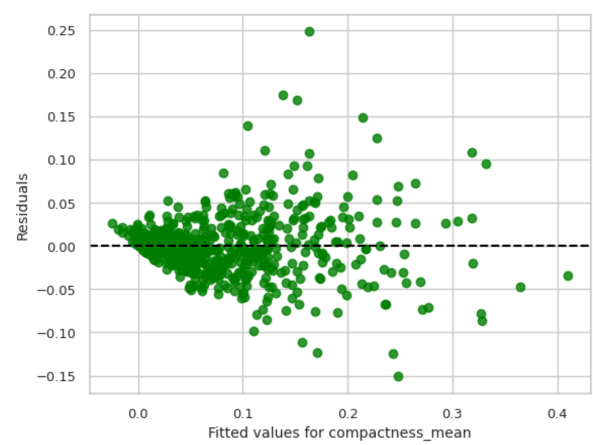


Figure 4: Residuals vs Fitted Values Plot

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          concavity_mean   R-squared:                       0.780
Model:                             OLS   Adj. R-squared:                  0.780
Method:                  Least Squares   F-statistic:                     2009.
Date:                 Mon, 03 Feb 2025   Prob (F-statistic):          1.62e-188
Time:                         01:41:32   Log-Likelihood:                 1062.9
No. Observations:                  569   AIC:                            -2122.
Df Residuals:                      567   BIC:                            -2113.
Df Model:                            1
Covariance Type:             nonrobust
====================================================================================
                       coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------
const               -0.0503      0.003    -14.463      0.000      -0.057      -0.043
compactness_mean     1.3331      0.030     44.823      0.000       1.275       1.391
==============================================================================
Omnibus:                       143.184   Durbin-Watson:                   1.915
Prob(Omnibus):                   0.000   Jarque-Bera (JB):              914.546
Skew:                            0.941   Prob(JB):                    2.56e-199
Kurtosis:                        8.919   Cond. No.                         19.2
==============================================================================
```

Figure 5: Summary of Ordinary Least Squares regression model

After computing the value, we have **r=0.883120670177251**. It indicates a strong linear positive relationship between concavity and compactness. An increase in compactness of the cell is generally accompanied by a proportional increase in concavity. As can be seen in Figure 1, data points are clustered around the line sloping upward, however, beyond a compactness value of 0.15, more scatter is observed, with dots moving further from the line, although the linear trend persists. Such strong correlation does not imply causation in any form however there is potential for a causal relationship between compactness and concavity, particularly given the established link between higher concavity and more irregular cell forms, which are concerning in cancer diagnosis. Future research could investigate these underlying biological mechanisms. One extraneous variable could be cell size. [1]

[1] #correlation: The correlation between compactness and concavity was calculated and the r-value suggested a strong positive linear relationship exists. Upon looking at the scatter plot, the reliability of the r-value was questioned due to heteroscedasticity, especially for higher values of compactness above 0.15. The difference between correlation and causation is explained and possible future direction of research was mentioned.

Now let's proceed with the coefficient of determination or R-squared which can be found in Figure 5. R-square is a precise measure in percentage which says whether it's practical to rely on a certain variable - in our case compactness - to predict another - concavity. In the table, R-squared is equal to **0.78**. Hence, about 78% of the total variation in concavity can be explained by the variation in compactness. It suggests that compactness is a reasonably effective predictor of concavity and can be a promising feature to use in diagnosis. But the remaining 22% of unexplained variation indicates that other factors may contribute to concavity that are not captured by this model.

To evaluate the reliability of the least square regression model and whether it is even fitting for this data, we have to check four key conditions. All conditions can be remembered as LINE and indicate: L - linear relationship variables; I - independence; N - normality of residuals; and E - equal variance (homoscedasticity).

To evaluate each of the conditions we will be relying on the 4 plots: Normality Probability (Q-Q) plot, Histogram of residuals, Scatter plot and Residuals vs. Fitted Values plot shown above. [2]

(L): The linear relationship based on the scatter plot appears to be strong, positive; pearson's r which is 0.88 indicate the same. However as we can see on the plot, the dots follow the shape of the cone rather than football shape, making dots more scattered with higher values of

[2] #dataviz: Four types of data visualization were used and interpreted appropriately to evaluate four key conditions for regression. In explanation, each plot provided aid in distinguishing a single condition while some plots obscured some of the information so multiple visualizations were made. The plots were used to connect with some of the claims made in text.

compactness. But a clear linear trend still persists so we can say that this condition is being fulfilled.

(I): Fine needle aspiration (FNA) which was used to collect the data samples cells from different areas of a tumor or from multiple patients, promoting independence. So we can say that this condition is being fulfilled.

(N): As we can see on the Normality QQ plot of residuals, residuals show a clear deviation from normality, especially in the tails, suggesting that the assumption of normality is violated. Histogram is not as sensitive to heavier tails or slight skewness as Normality QQ plot. The deviation from normality can be attributed to the skewness of the sample data itself. Regardless, it puts significant doubt on the reliability of this model.

(E): In Figure 4, the regression residuals are not evenly distributed. The plot shows that the data has heteroscedasticity. It poses a problem because ordinary least-squares (OLS) regression seeks to minimize residuals and in turn produce the smallest possible standard errors. OLS regression gives equal weight to all observations, but when heteroscedasticity is present, the cases with larger disturbances have more “pull” than other observations.

Considering the violation of a few conditions, the model is not very strong and might not be the best fit for this data. Possible improvements could include transforming the variables (e.g., log transformation) to stabilize variance and improve normality or exploring more advanced regression techniques such as weighted least squares. Additionally, other predictor variables might play a role in concavity beyond compactness, and a multivariate approach could provide a better explanation but won’t be explored in this paper.

The predictive equation is:

$$Concavity = -0.0503 + 1.3331 * Compactness \ (y = b0 + b1 * x)$$

For each unit increase in compactness, there is an increase of 1.3331 percentage points in concavity. If compactness is 0, we predict concavity to be -0.0503. However, the regression equation is not expected to make a reliable prediction near Compactness = 0 because this would require extrapolation beyond the range of given data.[3]

Significance testing[4]

Before proceeding, it's important to briefly cover the conditions for inference on slope. First 4 conditions were mentioned above. Finally, fifth one condition, randomness is being fulfilled as the data collected from Fine Needle Aspiration (FNA) is based on a random sample of cells from various areas of tumors and multiple patients, ensuring randomness in the sampling process.

[3] #regression: Ordinary Least Squares regression model was evaluated based on the explanatory power it had, quantified by R-squared in the context of the research. The regression equation was interpreted relying on the exact predictor and response variables in the study; coefficients and y-intercept were explained. Regression model was also evaluated based on the key 4 conditions which allowed us to arrive at the conclusion about the fitability of a particular model to explain variance in concavity relying on compactness.

[4] #significance: P-value was interpreted to explain whether we can reject or not the null hypothesis that there's no positive linear relationship between our key variables. The results of statistical significance analysis were reviewed and the implications of it in the context of study were considered. Step-by-step process is shown to demonstrate what lies in the foundation of p-value and how based on p-value we can make an inference about the population.

Study aims to identify whether we can infer from the sample that there's a strong linear positive relationship between concavity and compactness of the breast masses. First step of statistical significance analysis would be to construct clearly defined null and alternative hypotheses:

- H0: $\beta 1 = 0$
  - There's no linear relationship between concavity and compactness of the cell.
- HA: $\beta 1 > 0$
  - There's a positive linear relationship between concavity and compactness.

*$\beta 1$ - a slope of the population regression line that we infer about using estimates from the sample.

*One-tailed test is more suited for this case as the trend that was identified above was clearly positive; additionally, from outside research higher compactness is specifically associated with higher concavity.

Since neither of the statistical errors (Type I, Type II) seem to have severe negative consequences in the scope of this study, the significance level will be put to its default of 0.05. For analysis, p-value has to be calculated.

First, we need to compute the t-score. Using Python and comparing them to the results in Figure 5, we arrive at the following results: t-score = 44.823, with SE = 0.03. The t-score indicates how many standard deviations the observed value is from the null hypothesis mean. Our degrees of freedom are equal to n - 2. See Appendix D for the calculation in Python. The t-score of 44.823 results in a one-tailed p-value that is so small that it is shown as 0 in both table and python calculations.

This p-value represents the probability of observing values as extreme or more extreme, given that the null hypothesis is true. In the context of this study it would mean that if there's no linear relationship between concavity and compactness, the probability of observing the slope that was based on our sample data is nearing zero. It makes it reasonable then to reject the null hypothesis in favor of an alternative one, which indicates that there's a strong linear relationship between our variables.

Confidence interval[5]

To continue with our analysis, we can construct confidence intervals. These intervals will reflect the precision of our estimates for our slope. We set our confidence level to 95% because it provides a high level of certainty about the estimates.

The calculations were both done in Python and compared to those that can be seen in Figure 5. The full calculation is found in Appendix D. The resulting 95% confidence interval is [1.275, 1.391].

This means that, based on the frequentist interpretation of probability, if we were to repeatedly resample and calculate confidence intervals, we would expect the true population slope to fall within this interval in 95% of those cases. The 95% confidence interval for the slope [1.275, 1.391] does not include zero, supporting the conclusion from the p-value (which is near zero), further confirming the rejection of the null hypothesis.

---

[5] #confidenceintervals: Upper and lower values that are likely to contain the population parameter - the slope of the regression line was calculated. The choice of the confidence interval of 95% was justified. The interval was interpreted correctly and is used to inform the conclusion about alternative hypotheses as lack of overlap with null hypothesis was observed. The results matched with what was expected after significance testing emphasizing the tight connection between two.

Conclusion

The paper explores the relationship between breast cell compactness and concavity using a sample of data from Breast Wisconsin Dataset. The study found a strong positive correlation ($r = 0.883$) between these variables, and the regression model that was used was able to explain 78% of the variation in concavity relying on compactness. However, some conditions were violated which put a doubt on whether the model can be a good fit for the data. Despite these limitations, the significance testing (p-value near zero) and confidence interval ([1.275, 1.391]) support rejecting the null hypothesis, indicating a significant positive relationship between compactness and concavity. Additionally, when conducting an inductive process we have to emphasize that the inference about the population is likely to be true given the results however there's always room for errors. This suggests compactness could potentially predict concavity in breast cancer diagnosis. However, the model's limitations and unexplained variance highlight the need for cautious interpretation and further research, possibly incorporating additional variables or advanced statistical techniques to improve predictive accuracy of the regression model that was presented in the study.[6][7][8]

[6] #induction: The conclusion serves as a strong example of induction as it makes the generalization about the population. The linear relationship between concavity and compactness was explored and whether this linear relationship is strong enough in population that it can be used for practical purposes. The conclusion emphasizes the strength and reliability of the induction by mentioning the statistical tools used and results from them. It goes further by outlining the limitations of generalization that are always present in induction.

[7] #organization: The study has an easy-to-follow format where each part is being built on top of each other. The titles are used for easier navigation.

[8] #professionalism: The assignment follows the established guidelines by Minerva University. The tone of the paper is formal. The AI statement showed my reflection on the assignment itself and the learning it brought to me. The reflections were written with an intention to gain more knowledge and engage more with material.

Reflections

Testing: To confirm the results I have received from my computations, I have used the approach of checking the results by performing manual calculations using an online calculator, desmos. After getting results from manual calculations, I've compared them with values I got from Python computations. The results matched which proved that my calculations were correct. Additionally, instead of relying completely on the results that were provided by regressionmodel.summary() package, Python was used separately to perform calculations step by step following the formulas from the readings

Acknowledgements

I would like to acknowledge some of the sources that I relied on when constructing my introduction as well as evaluating some of the conditions for the model. Here are the links to those sources:

https://www.nature.com/articles/s41598-021-03813-8

https://www.cdc.gov/breast-cancer/statistics/index.html

https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data

Additionally, I would like to acknowledge my FA professor, prof.Teranna who through sessions allowed me to deepen my understanding and interpretation of some of the fundamental concepts in this unit. Also, I am truly grateful to my classmates as their questions in the past sessions

pushed me to think deeper about some of the concepts taught as well as my roommate who helped me to make a decision how to proceed with some conditions being violated for my model.

AI statement

In this assignment, I used AI only for the purpose of debugging the code. I utilized an AI assistant that is being offered by Google Colab. Additionally, Perplexity AI was used to search for some sources when writing an introduction for the paper. Grammarly and Wordtune were used for sentence level editing and overall organization of some paragraphs.

Appendix A

+ Код + Текст

```
[1] '''

    All used libraries were imported in this cell.

    '''

    import statsmodels.api as sm
    import numpy as np
    import pandas as pd
    import matplotlib.pyplot as plt
    import seaborn as sns
    from scipy import stats
    import statsmodels.api as statsmodels
```

Data prep

```
[2] df = pd.read_csv("https://course-resources.minerva.edu/uploaded_files/mu/00353110-6347/breast-cancer-wisconsin.csv")
    df.head()
    df = df.drop(columns=['id'], inplace=False)
    df.head()
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean	symmetry_mean	...	texture_worst	perimeter_worst	area_worst	smoothness_worst	compactness_worst	concavity_worst	point
0	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	...	17.33	184.60	2019.0	0.1622	0.6656	0.7119	
1	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	...	23.41	158.80	1956.0	0.1238	0.1866	0.2416	
2	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	...	25.53	152.50	1709.0	0.1444	0.4245	0.4504	
3	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	...	26.50	98.87	567.7	0.2098	0.8663	0.6869	
4	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	...	16.67	152.20	1575.0	0.1374	0.2050	0.4000	

5 rows × 32 columns

Двічі натисніть клітинку, щоб змінити її (або натисніть клавішу Enter)

```
[3] units = {"area mean": "square pixels", "symmetry mean": "unitess", "concavity mean": "unitless", "compactness mean":"unitless", "smoothness mean":"unitless", "perimeter mean": "pixels"}
```

```
[3] units = {"area_mean": "square pixels", "symmetry_mean": "unitess", "concavity_mean": "unitless", "compactness_mean":"unitless", "smoothness_mean":"unitless", "perimeter_mean": "pixels"}
```

```
[4] if df['concavity_mean'].isnull().values.any():
         print("Given column has NaN values.")
    else:
        print("Given column does not have NaN values.")
```

```
Given column does not have NaN values.
```

```
[5] if df['compactness_mean'].isnull().values.any():
         print("Given column has NaN values.")
    else:
        print("Given column does not have NaN values.")
```

```
Given column does not have NaN values.
```

Appendix B

```python
The code was adapted from "Decribing Data" assignment.

'''


def my_mean(data): #passes the list or array of numerical values
    sumtotal = 0
    n = len(data)


    for element in data: #iterates over each element in the dataset
        sumtotal += element


    return sumtotal/n #mean formula


def my_median(data):

    #sort the values in the dataset from the smallest to the greatest
    sorted_data = sorted(data)
    n = len(sorted_data)


    if n % 2 == 1: #odd number of elemements
        return sorted_data[n//2] #index of the middle element

    else: #even number of elements
        middle_1, middle_2 = n//2 - 1, n//2 #"n//2 - 1" index of the first middle element, " n//2" index of the second middle element.
        return (sorted_data[middle_1] + sorted_data[middle_2]) / 2 #average of two middle elements



def my_count(data, item):

    occurences = 0
    for element in data: #iteratation over each element in the dataset
        if element == item:
            occurences += 1
    return occurences



def my_mode(data):
    max_count = 0


    max_items = [] #the empty list where mode (or modes) will be stored
```

```python
    max_items = [] #the empty list where mode (or modes) will be stored

    for element in data:
        current_count = my_count(data, element) #counts number of the same element in the dataset.

        if current_count > max_count:
            max_count = current_count
            max_items = [element]
        elif current_count == max_count and element not in max_items:
            max_items.append(element)


    if max_count == 1:
        return None
    return max_items if len(max_items) > 1 else max_items[0]


def my_sample_SD(data):
    sum_of_squares = 0
    n = len(data)
    mean = my_mean(data) #mean is part of standart deviation formula


    for element in data:
        sum_of_squares += (element - mean)**2

    return (sum_of_squares/(n-1))**0.5 #final operations for std formula


def my_range(data):


    sorted_data = sorted(data)
    return sorted_data[-1] - sorted_data[0] #indexing that extracts last and first value in the dataset


def my_skewness(data):

   n = len(data)
   if n < 3:
        raise ValueError("Skewness requires at least 3 data points.")

   mean = my_mean(data)
   std = my_sample_SD(data)

    # Calculate skewness using algebraic formula
   skewness = (n / ((n - 1) * (n - 2))) * np.sum(((data - mean) / std) ** 3)

   return skewness
```

$$Skewness = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left(\frac{x_i - mean}{Stdev} \right)^3$$

```
concavity = df['concavity_mean']
compactness = df['compactness_mean']
```

```
count_concavity = len(concavity)
count_compactness = len(compactness)

print("Concavity Count:", count_concavity)
print("Compactness Count:", count_compactness)
```

```
Concavity Count: 569
Compactness Count: 569
```

```
statistics = {
        'Concavity': {
            'Mean': my_mean(concavity),
            'Median': my_median(concavity),
            'Mode': my_mode(concavity),
            'Standard Deviation': my_sample_SD(concavity),
            'Range': my_range(concavity),
            'Skewness': my_skewness(concavity),

        },
        'Compactness': {
            'Mean': my_mean(compactness),
            'Median': my_median(compactness),
            'Mode': my_mode(compactness),
            'Standard Deviation': my_sample_SD(compactness),
            'Range': my_range(compactness),
            'Skewness': my_skewness(compactness),
        }
    }


print(statistics)
```

```
54, 'Mode': 0.0, 'Standard Deviation': 0.0797198087078935, 'Range': 0.4268, 'Skewness': 1.40
```

Appendix C

```
The code was adapated from FA51 Session 2 - (1.2) Quantifying Correlation / January 15th, 2025

'''

def corr_scatter(column_x, column_y):

    r = df[column_x].corr(df[column_y],method='pearson')
    return r
```

```
print("Pearson's R is equal to:", corr_scatter('compactness_mean','concavity_mean'))
```

```
Pearson's R is equal to: 0.883120670177251
```

```
'''


The code was adapted from FA51 Session 5 - (3.1) Synthesis: Correlation, Regression, and Statistics / January 27th, 2025


'''




def mult_regression(column_x, column_y):

    if len(column_x)==1:
        plt.figure()
        sns.regplot(x=column_x[0], y=column_y, data=df, marker="+",fit_reg=True,color='orange', ci=None)
        plt.xlabel(column_x[0] + "\n[" + units[column_x[0]] + "]")
        plt.ylabel(column_y + "\n[" + units[column_y] + "]")

    # define predictors X and response Y:
    X = df[column_x]
    X = statsmodels.add_constant(X)
    Y = df[column_y]

    # construct model:
    global regressionmodel
    regression_model = statsmodels.OLS(Y,X).fit() # OLS = "ordinary least squares"
    print(regression_model.summary())


    residuals = regression_model.resid

    # QQ plot to check the normaility condition
    sm.qqplot(residuals,fit=True, line='45')
    plt.show()
```

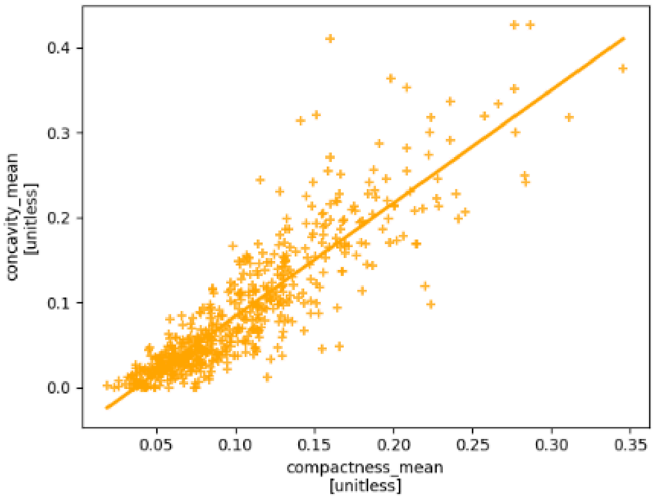
```
# Residuals Histogram
sns.histplot(residuals, kde=True, color="blue")
plt.xlabel("Residuals")
plt.ylabel("Frequency")
plt.show()

# Residuals vs Fitted Values Plot
sns.residplot(x=regression_model.fittedvalues, y=residuals, color='green')
plt.xlabel("Fitted values for compactness_mean")
plt.ylabel("Residuals")
plt.axhline(y=0, color='black', linestyle='dashed')  # Reference line at 0
plt.show()
```

```
mult_regression(['compactness_mean'],'concavity_mean')
```

```
==============================================================================
mnibus:                       143.184   Durbin-Watson:                   1.915
rob(Omnibus):                   0.000   Jarque-Bera (JB):              914.546
kew:                            0.941   Prob(JB):                    2.56e-199
urtosis:                        8.919   Cond. No.                         19.2
==============================================================================

otes:
1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```



Appendix D

```
'''

The code was adapted from FA51 Session 5 - (3.1) Synthesis: Correlation, Regression, and Statistics / January 27th, 2025

'''




r = 0.883120670177251
x_mean = 0.10434098418277686
y_mean = 0.08879931581722322
sx = 0.0528127579325122
sy = 0.0797198087078935
n = 569

b1 = r * (sy / sx)
print("b1 =",round(b1,3))

SE = (sy / sx) * ((1 - r**2)/ (n- 2))**0.5 #we can compare results with ones we have in the table
print("SE =",round(SE,3))

t95 = stats.t.ppf(0.975,n-2)
print("t95 =",round(t95,3))

lower_bound = b1 - t95*SE
upper_bound = b1 + t95*SE

print("interval =", [round(lower_bound,3),round(upper_bound,3)])
```

```
b1 = 1.333
SE = 0.03
t95 = 1.964
interval = [1.275, 1.391]
```

```
t_statistic = b1 / (SE) #we can compare the results with table above
print("t-statistic =",round(t_statistic,3))
```

```
t-statistic = 44.823
```

```
p_value = 1 - stats.t.cdf(t_statistic, df=n - 2)  #one-tailed test

print("p-value:", p_value)
```

```
p-value: 0.0
```